

When Diamonds Are No Longer Precious

Diamonds are precious stones, but this may end within the decade. Synthetic diamonds have been around since the 1950s, but have always been very small—far too small to be gemstones in jewellery. Instead they found countless uses in industry such as for abrasives, cutting edges and as heat sinks for high-power semiconductor lasers. New techniques in recent years, however, are making synthetic diamonds bigger. In some cases, a lot bigger. And they are making them quickly. Chemical & Engineering News reported in its February 2, 2004 issue that a company named Gemesis in Sarasota, Florida USA, can grow a synthetic diamond of 2.8 carats in less than four days

The World Future Society, in its publication *Twenty Forecasts for 2011-2025*, wrote: “Breakthrough processes are creating laboratory-grown diamonds that are molecularly identical to natural diamonds, yet a fraction of the cost. As large inexpensive luxury diamonds become commonplace, consumers may turn to other things with more cachet.”

One of the earliest public warnings of this trend appeared in *Wired Magazine* back in 2003. The article titled *The New Diamond Age*, written by Joshua Davis, was subtitled: “Armed with inexpensive, mass-produced gems, two startups are launching an assault on the De Beers cartel.” This article opened with Mister Davis travelling to Antwerp Belgium and handing a gem dealer a synthetic diamond grown by Gemesis for appraisal. It cost Gemesis about a thousand dollars to make it. The dealer, who was not told its origin, valued it between ten and fifteen thousand dollars.

Manufacturing jewellery-sized diamonds at ever decreasing prices is a trend driven by two powerful forces: technological innovation and the desire to make money by undercutting the competition. Two forces which are unlikely to vanish.

The largest supplier of natural diamonds on earth, the De Beers cartel, is so frightened by this that they have spent a great deal of money to develop devices which can tell a natural diamond from a synthetic diamond,

and are offering these devices to large gemological labs at no cost. But the devices, DiamondView and DiamondSure, have shown themselves unreliable in independent testing such as that done by Marty Haske, president of Adamas Gemological Laboratories, and by the Gemological Institute of America.

This lack of reliability should not be surprising since synthetic diamonds are composed of the same substance as natural diamonds: carbon atoms arranged in a type of cubic lattice called diamond cubic. This is why natural diamonds are not called “real” diamonds. They can’t be. They are no more real than those grown synthetically. To make such a claim would be the same as insisting that the ice cubes in your freezer are fake ice

and only the ice found naturally frozen outside on a winter’s day is real.

Retailers and wholesalers in every phase of the natural diamond supply chain understand the danger synthetic diamonds pose to their business. Blue Nile, Inc., for example, the largest online retailer of fine jewelry reported on page 15 of its 2011 annual report: “...attempts have been made to develop and market synthetic stones and gems to compete in the market for diamonds and diamond jewellery. We expect such efforts to continue in the future. If any such efforts are successful in creating widespread demand for alternative diamond products, demand and price levels for our products would decline and our business and results of operations would be substantially harmed.”

Already there are online retailers specialising in synthetic diamonds. It’s all they sell. And they do not present themselves as cheap diamond discounters, but as a source of high fashion for shoppers of sophistication. They merely offer a more reasonable entry to into the world of high end jewellery.

As the technology of growing synthetic diamonds improves—producing stones both cheaper and larger—the price gap between natural and synthetic diamonds will get progressively wider each year. And as more retailers offer them to consumers, a growing portion of the gem buying public may be lured away. If demand for the far more expensive natural stones declines, so too will their prices.

Whether this financial impact will manifest itself in one year or ten years is uncertain, but in 2007 the Gemological Institute of America began grading synthetic diamonds for carat, colour, clarity and cut—just as they do for natural diamonds—which gives additional validity to synthetic diamonds and the manufacturers which grow them. And in some people’s eyes place the synthetic on the same level as the natural.

Individuals who own diamonds today, or who have invested in diamonds hoping that their investments will remain safe or provide a hedge against inflation, need to be aware of these trends and decide if this asset class is right for their future. **[1]**



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